

# **IIoT Network Analysis: Age of Information and Reliability Trade-offs**

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## **Conceptual Understanding**

Age of Information (AoI) is a modern networking metric that characterizes the "freshness" of data from the perspective of the receiver. Unlike traditional latency, which measures the time a specific packet takes to travel from point A to point B, AoI measures the time elapsed since the most recently received update was generated at the source. In Industrial Internet of Things (IIoT) applications, AoI is critical because many automated systems rely on current state information to make safe and efficient decisions. For instance, in a smart warehouse utilizing Autonomous Guided Vehicles (AGVs), the central controller needs the most recent positional data of every robot to prevent collisions. If the AoI is high, the controller is essentially working with "stale" data, which could lead to an AGV being commanded to move into a space that is no longer clear.

The distinction between AoI-oriented traffic and deadline-oriented traffic is fundamental to IIoT network design. AoI-oriented traffic is focused on maintaining a continuous stream of fresh status updates where the occasional loss of a packet is acceptable, provided a newer update arrives shortly after to "reset" the age. An example of this is a temperature sensor in a chemical plant that sends updates every second; the system needs to know the current heat level, but missing one specific packet is fine if the next one arrives on time. Conversely, deadline-oriented traffic is mission-critical data that must arrive within a strict timeframe to be useful at all. A real-world example would be an emergency "kill-switch" signal triggered by a hardware failure. If that specific signal does not arrive within its 10ms deadline, the machine could sustain permanent damage, regardless of how "fresh" any subsequent data might be.

## **Data Exploration and Visualization**

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## Data Exploration and Visualization

Upon loading the IIoT network dataset, the initial exploration revealed a complex environment characterized by high variance in the Age of Information. The dataset contains several features including `transmission_probability`, `capture_threshold`, `num_nodes`, and `channel_quality`.

Summary statistics indicated that the `age_of_information` column contained extremely large values and mathematical infinities, which represent instances where communication failed entirely and the data became infinitely stale. This necessitated a rigorous cleaning phase where infinite values were treated as missing data to allow for accurate mathematical modeling.

The visualizations derived from the dataset highlight three distinct patterns that define the relationship between network parameters and performance:

1. **The Transmission Probability Stability:** In the scatter plot of Transmission Probability vs. Age of Information, we observe that most data points cluster near a low AoI across all probability levels. However, extreme outliers reaching AoI values over 40,000 occur more frequently at lower transmission probabilities (around 0.1). As the transmission probability increases toward 1.0, the density of extreme outliers seems to decrease, suggesting that while higher frequency doesn't guarantee a lower average age, it can reduce the likelihood of the system remaining in a "stale" state for extended periods.
2. **Traffic Type Distribution and Outliers:** The box plot of Age of Information Grouped by Traffic Type reveals that both deadline-oriented and AoI-oriented traffic maintain very low median AoI values, appearing almost flat on the scale. However, both groups are plagued by significant outliers. Notably, the AoI-oriented traffic shows a higher density of these outliers, with some reaching above 50,000. This suggests that while the protocol

handles continuous updates well on average, it is susceptible to severe "aging" spikes that could disrupt industrial monitoring.

3. **Network Reliability Correlations:** The Correlation Heatmap provides a clearer picture of what actually drives network failure. While AoI shows very weak correlations with individual parameters, Packet Loss Probability (PLP) is strongly influenced by the physical network environment. There is a significant positive correlation (0.54) between `num_nodes` and `packet_loss_probability`, and a strong negative correlation (-0.45) between `channel_quality` and PLP. This indicates that reliability is primarily a function of congestion and signal integrity, which indirectly causes the massive AoI spikes observed in the other plots.

## **Machine Learning Model Development**

The Random Forest Regressor was developed using `transmission_probability`, `capture_threshold`, `num_nodes`, and `channel_quality` as features. After scaling the data to normalize the influence of different units, the model was trained to predict the AoI. The model's performance yielded a Mean Squared Error (MSE) of 3,423,000 and a negative R-squared score. In a professional context, these results indicate that the model struggled to generalize across the extreme variance of the dataset, particularly when dealing with the high-age outliers that remained after cleaning. Despite the high error, the feature importance analysis was highly instructive, showing that `capture_threshold` (0.4401) and `num_nodes` (0.3152) were the most significant predictors of AoI. This tells us that the physical capability of the receiver to "capture" a signal amidst noise and the sheer number of competing devices are more influential to data freshness than how often an individual device tries to talk.

When the trained Random Forest model was used to predict AoI for hypothetical network configurations, the results aligned with our engineering intuition. For example, a configuration with high transmission probability (0.9), a high number of nodes (7.0), and poor channel quality (0.4) resulted in a predicted AoI that reflected a congested and struggling network. Interestingly, the model predicted an AoI of 16.05 for a moderate setup, demonstrating that as we increase the number of nodes, the system enters a state of diminishing returns. These predictions reinforce the idea that IIoT network parameters are deeply interconnected; you cannot simply maximize one parameter without negatively impacting the overall system state.

## **Analysis and Insights**

The primary factor influencing the AoI-PLP trade-off is the balance between update frequency and channel congestion. High transmission probabilities are intended to keep data fresh, but they directly increase the probability of packet collisions, which in turn increases PLP. When PLP rises, the AoI inevitably spikes because the "fresh" updates are being dropped before they reach the destination. To optimize this, I propose two key strategies. First, implementing an Adaptive Transmission Scheme where nodes reduce their transmission probability when high PLP is detected would allow the network to "breathe" and recover from congestion. Second, the use of Traffic Slicing can ensure that critical deadline-oriented packets are given a dedicated "fast lane," preventing them from being caught in the congestion created by high-volume AoI-oriented status updates.

The insights gained here have direct applications in high-stakes IIoT environments. In remote robotic surgery or teleoperation, understanding the AoI-PLP trade-off is vital for ensuring that the visual feedback the surgeon sees is fresh enough to act upon, while the control signals remain

100% reliable. Similarly, in smart grid management, utility providers can use these models to determine the optimal density of smart meters. By balancing the update frequency of power consumption data with the reliability of the local mesh network, they can ensure the grid responds to load changes in real-time without crashing the communication infrastructure during peak hours.

### **Bonus Challenge: Deep Learning Comparison**

The implementation of a multi-output Deep Learning model using TensorFlow provided a more robust alternative to the Random Forest approach. By utilizing a neural network with hidden layers of 64 and 32 neurons, the model was able to predict both AoI and PLP simultaneously. The Deep Learning model achieved a significantly lower MSE for AoI (2,163,060) compared to the Random Forest (3,423,000). This improvement is likely due to the neural network's ability to use "shared representation learning," where the model learns the underlying physics of the network—such as how nodes and interference interact—and applies that knowledge to predict both freshness and reliability at once. While the Random Forest is better for identifying which specific features are most important to an engineer, the Deep Learning model proved to be the more accurate "black box" for complex, multi-objective predictions in a volatile IIoT landscape.